

FUKUI, Japan

WMO: 476160

Lat: 36.0554N

Lon: 136.2230E

Elev: 17

StdP: 101.12

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	-0.9	-6.4	2.2	4.7	-5.1	2.5	4.2	8.4	3.6	7.6	3.8	2.5	190	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	34.2	24.9	33.2	24.7	32.0	24.4	26.1	31.4	25.6	30.7	25.1	30.0	4.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	19.8	28.3	24.2	19.1	27.9	23.7	18.5	27.6	81.0	31.3	78.7	30.8	76.7	30.1	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.0	6.1	DB	-3.4	36.0	1.0	0.8	-4.1	36.6	-4.7	37.1	-5.3	37.5	-6.0	38.1	(4)
(5)			WB	-4.0	27.0	0.9	0.7	-4.7	27.5	-5.3	27.9	-5.8	28.2	-6.5	28.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	3.3	3.9	7.3	13.0	18.3	22.2	26.6	27.9	23.5	17.6	11.5	6.1	(6)	
	DBStd	8.92	2.18	2.64	3.31	3.65	2.89	2.43	2.55	2.20	2.97	2.92	3.14	2.93	(7)	
	HDD10.0	634	206	173	97	13	0	0	0	0	0	0	19	126	(8)	
	HDD18.3	2044	464	405	341	164	36	1	0	0	1	48	205	379	(9)	
	CDD10.0	2525	0	1	14	102	259	367	513	556	406	237	65	5	(10)	
	CDD18.3	893	0	0	0	3	36	118	255	297	156	26	1	0	(11)	
	CDH23.3	7842	0	0	0	32	240	666	2448	3256	1095	104	1	0	(12)	
	CDH26.7	2921	0	0	0	4	38	158	971	1401	336	13	0	0	(13)	
(14) Wind	WSAvg	2.8	2.6	2.7	3.0	3.1	3.1	2.6	2.7	2.8	2.7	2.6	2.6	2.6	(14)	
(15) Precipitation	PrecAvg	2253	279	165	154	131	143	154	229	146	202	150	203	296	(15)	
	PrecMax	2716	465	260	237	217	377	287	586	358	342	249	406	665	(16)	
	PrecMin	1537	96	49	72	46	57	40	53	24	65	80	117	118	(17)	
	PrecStd	275	94	51	46	46	76	59	130	80	97	44	55	125	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	16.0	21.0	26.7	29.6	31.7	35.2	35.7	33.6	28.2	22.6	17.3	(19)	
		MCWB	8.1	10.4	12.2	15.6	18.6	22.5	25.2	25.3	24.1	21.2	15.7	12.4	(20)	
	2%	DB	10.0	12.7	18.1	23.7	27.7	30.0	34.0	34.6	31.9	25.9	20.3	14.5	(21)	
		MCWB	6.8	8.3	10.9	14.8	17.5	22.1	24.8	25.0	23.9	19.4	14.9	10.6	(22)	
	5%	DB	8.5	10.6	15.7	21.7	26.0	28.5	32.9	33.6	30.2	24.2	18.5	12.5	(23)	
		MCWB	5.8	6.9	10.0	14.0	16.8	21.0	24.7	25.0	23.3	18.4	14.1	9.7	(24)	
	10%	DB	7.1	8.7	13.5	19.7	24.3	27.0	31.5	32.5	28.5	22.7	16.9	10.9	(25)	
		MCWB	4.9	5.8	9.3	13.0	16.4	20.5	24.3	24.8	22.5	17.6	13.5	8.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	12.1	14.0	19.0	21.4	24.8	26.6	27.1	25.6	22.4	17.7	13.9	(27)	
		MCDB	11.2	15.0	17.5	22.5	24.9	29.1	32.5	32.6	30.7	26.1	19.9	15.7	(28)	
	2%	WB	7.4	9.2	12.3	16.4	19.8	23.7	25.7	26.3	24.7	20.9	16.2	11.6	(29)	
		MCDB	9.2	11.8	16.1	21.0	24.4	27.5	31.3	31.6	29.3	24.4	18.8	13.6	(30)	
	5%	WB	6.1	7.5	11.0	15.2	18.5	22.7	25.2	25.8	24.1	19.7	15.1	9.9	(31)	
		MCDB	8.0	9.9	14.5	19.6	23.1	26.4	30.6	31.0	28.5	22.5	17.6	12.2	(32)	
	10%	WB	5.0	6.0	9.6	14.0	17.6	21.7	24.7	25.4	23.4	18.6	13.9	8.6	(33)	
		MCDB	6.9	8.4	13.0	18.0	22.2	25.3	29.8	30.3	27.5	21.7	16.3	10.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.7	8.7	9.7	9.1	7.6	7.1	7.8	7.6	8.4	7.9	6.4	(35)	
		MCDBR	8.6	10.8	12.9	13.3	12.7	10.2	9.3	9.2	9.2	10.0	10.4	9.6	(36)	
	5% WB	MCWBR	5.6	6.9	7.2	6.1	5.1	3.9	3.0	2.9	3.3	4.5	5.9	6.2	(37)	
		MCWBR	7.7	9.6	10.4	9.9	9.3	7.5	8.1	8.0	7.5	8.1	8.1	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.435	0.464	0.528	0.550	0.551	0.573	0.567	0.539	0.486	0.452	0.434	0.435	(40)	
		taud	1.979	1.933	1.824	1.826	1.883	1.908	1.983	2.061	2.167	2.154	2.107	2.010	(41)	
	Ebn at Noon	Ebn at Noon	714	749	739	750	757	739	739	751	772	758	716	684	(42)	
		Edh at Noon	138	162	198	208	199	194	179	162	139	129	121	127	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.50	3.66	4.66	5.22	4.94	4.88	5.09	3.96	3.17	2.17	1.55	(44)	
	RadStd	RadStd	0.14	0.24	0.30	0.40	0.56	0.41	0.71	0.54	0.33	0.27	0.18	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	+0.45	+0.65	N/A	N/A	N/A	N/A	
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.26	N/A	N/A	N/A	N/A	

Nomenclature: See separate page